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FORMAL EPISTEMOLOGY • GAME THEORY • FOUNDATIONS OF PHYSICS

Populations and Strategy in Cognitive Spaces:

Lag, Horizons, Finite Computation and the Gambler

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Formalisation developed with Claude, an AI system by Anthropic (see Author contributions)

Abstract. This is the population and strategy sequel to a core framework (Paper B of this series) in which a cognitive space is a Turing ideal, agents climb by uncertified promotion, and encounters are merger, elimination or trade; it uses the rung-distance geometry of a companion paper. Run forward in time, a population under canonical ascent shows no lateral drift, only one-sided vertical recession at the rate difference, and every canonical lineage tends to the single ideal point unless a lineage of unbounded lag exists, so the geometry's boundary question is whether a population has one horizon or several. A cosmological constant makes total computation finite, so every physical agent is at rung 0 for ever; we define the *settled profile*, a resource-bounded notion of rung, and show that an agent cannot compute its own. Physical readings develop the spectral gap as a conditional certificate of convergence and read Malament–Hogarth horizons as purchased certificates. We then give the information structure of the strategic game: no universal detector of superiority exists, bluff-proof detectors convince only a meagre class, randomness demonstrates strict superiority in the limit while genericity cannot, and subsumption is demonstrable with a witness or from one rung up, but by no protocol at the prover's own rung. Under two stated postulates conquest is sterile: certified growth is by consent. Finally the *Gambler*, a strategic agent (C, P, r) whose source r it does not hold, is unpredictable even when cognitively subsumed and is the only dynamics of a rational population. Separating prediction from audit, a random source is recovered in hindsight three rungs up with a certified delay, a delay-coded generic source can evade any prescribed finite rung, and neither is ever predicted. Demonstration spends concealment; the source buys unreadability but no height.

Series note. This is Paper C of the series *Rungs, Voids and Effective Closure*, the first programme of the study *Effective Horizons and Cognitive Spaces* (synopsis in [1]), four working papers on a model of agents whose cognitive reach is a Turing ideal and who extend it by uncertified convergence. The core paper [2] supplies the order of reachability, effective closure and the arithmetical ladder, the agent and encounter model, and the synthesis of the programme; the geometry paper [3] turns closure steps into a distance and proves, among other things, that the delays at which hidden content becomes recoverable can escape every rung of domination. Everything here is downstream of those two: time, populations and finite resources put the geometry on a clock, and strategy asks what an agent can predict, demonstrate or audit about another; the points at which this paper attaches to the core are listed in [2, Section 12]. The fourth paper [4] takes the rate dynamics of Section 2 into a quantitative model and applies it to the Fermi problem. The thread is the one the synthesis

in [2, Section 13] names: limits determine what may become true, and moduli and channels determine what can become usable.

Summary of results. Section 1 records what is imported. Section 2 runs a population forward: Proposition 1 bounds lag under promotion (linear at most, bounded for equal-rate canonical climbers, one-sided between a climber and a stationary agent), and from it follow the absence of lateral drift under canonical ascent, the identification of the one-or-several-horizons question of [3] with a population question, the three pair regimes (convergence, fellow-travelling, vertical and lateral divergence), a labelled-population ontology, and the freeze-out that a cosmological constant imposes. Section 3 defines the realised trace and the settled profile (Definition 1) and proves, for canonical promotions, that profiles settle in layers and that an agent cannot compute its own (Proposition 2). Section 4 gives the physical readings: the spectral gap as a conditional dynamical modulus, the undecidability of the gap [6] and size-driven transitions [5] as the two impossibilities of [2], and Malament–Hogarth horizons as purchased certificates. Section 5 contains the paper’s main theorems: the four facts about detectors of superiority (Theorem 3) and the three-part theorem on demonstrating subsumption (Theorem 4); under Postulates 1 and 2 the sterility of conquest (Proposition 3, Corollary 1); the strategic agent and the Gambler (Definition 3) with Propositions 4–6, of which the last two are consequences in prose under the postulates rather than theorems in a specified game; and the audit results, Propositions 7–9, built on the hidden-information theorem of [3], with their consequences for time scales, concealment and the audit horizon. Section 6 lists the open problems and Section 7 closes. The quantitative rate model, the percolation conjecture in its accessibility form and the Fermi function that this dynamics supports are developed in [4]; nothing here is needed by [2]. The paper is speculative by design: its theorems are short and its postulates explicit, but the model choices (elimination requires a model; lexicographic preferences) are choices, and which claims are theorems and which hold under postulates is marked throughout.

1 IMPORTED DEFINITIONS AND RESULTS

We use the following from [2] and [3] without reproof. An *agent* is a pair (C, P) of Turing ideals, its certified and committed spaces; agents are individuated by their pairs, so a *population* $\mathcal{P}$ is a finite set of degrees ([2, Def. 4, Remark 5]). An agent *promotes* by committing to a limit of its own approximations; after k promotions from x its committed space is at most the ideal below $x^{(k)}$ ([2, Prop. 5]). Content passes between agents as algorithm, as method, or as testimony ([2, Prop. 6]). An *encounter* is a merger $\mathcal{P} \mapsto (\mathcal{P} \setminus \{x, y\}) \cup \{x \oplus y\}$, which settles $x \oplus y$; an elimination $\mathcal{P} \mapsto \mathcal{P} \setminus \{y\}$; or a trade, in which each promotes some of the other’s content ([2, Def. 7]). The *rung distance* is $d(x, y) = \max(\delta(x, y), \delta(y, x))$ with $\delta(x, y)$ the least k with $y \leq_T x^{(k)}$; $h(x) = d(x, \emptyset)$; a *generalised-low agent* has $x^{(k)} \equiv_T x \oplus \emptyset^{(k)}$; $e_k(x)$ and $c_k(x)$ are the highness and fellow-travelling quantities of [3].

Theorem 1 (Imported from [2]). Let $I_x = \{y : y \leq_T x\}$. (1) An observer computable from x' verifies I_x in the limit. (2) No observer, of any power, verifies $2^{\mathbb{N}} \setminus I_x$ in the limit. Also ([2, Thm. 5]): the halting problem is settled by a merger of two agents neither of whom settles it, and every noncomputable c.e. space is the merger of two strictly smaller ones.

Theorem 2 (Imported from [3]). (1) d is an extended metric, finite exactly between mutually arithmetical sets, and $|h(x) - h(y)| \leq d(x, y) \leq \max(h(x), h(y))$. (2) Every canonical ascent $x, x', \dots$ is a geodesic ray asymptotic to the trunk, at distance $c_t(x) \leq h(x)$ from $\emptyset^{(t)}$ at step t ; all such rays determine one ideal point ξ ; intermediate c.e. degrees give rays at distance exactly one from the trunk for ever. (3) Every geodesic ray r has $h(r(i)) = i + O(1)$, and the boundary at infinity is a single point iff every ray has bounded lag $\delta(r(i), \emptyset^{(i)})$. (4) The ultrametric $D(x, y) = \min\{k : x \oplus \emptyset^{(k)} \equiv_T y \oplus \emptyset^{(k)}\}$ satisfies $d \leq D \leq d + \max(\varepsilon(x), \varepsilon(y))$ for the excess ε , zero exactly on generalised-low agents. (5) For two mutually random generalised-low agents a, b of height H with $e_k(a) = e_k(b) = 0$, $d(a \oplus \emptyset^{(s)}, b \oplus \emptyset^{(t)}) = \max(s, t, H) - \min(s, t)$; $(\mathcal{A}, d)$ is not a graph metric and is not Gromov-hyperbolic in the four-point sense.

2 COSMIC TIME: WHAT THE MODEL FORCES

Run the population forward, each agent promoting at its own rate. Write x_t for agent x after t steps.

Proposition 1 (Lag under promotion). **1.** Under any promotion policy, $h(x_t) \leq h(x_0) + t$, and hence $d(x_t, y_t) \leq \max(h(x_0), h(y_0)) + t$: lag grows at most linearly.

- 2.** If both agents promote canonically, $d(x_t, y_t) \leq c_t(x_0) + c_t(y_0) \leq h(x_0) + h(y_0)$ for all t : equal-rate canonical climbers never separate.
- 3.** If x does not promote and y promotes canonically from a generalised-low y_0 with $e_k(y_0) = 0$ and $d(x, y_0) = H$, then $\delta(y_t, x) \rightarrow 0$ while $\delta(x, y_t) = t$ for $t \geq H$: the faster agent sees the slower completely after finitely many steps, and the slower needs t rungs to see the faster.

Proof. (1) Promotion adds content below the current jump, so h rises by at most one per step; then the sandwich bound $d(x, y) \leq \max(h(x), h(y))$ of [3]. (2) Both $x_0^{(t)}$ and $y_0^{(t)}$ are within $c_t(\cdot)$ of $\emptyset^{(t)}$ by Theorem 2(2) and [3], and c_t is non-increasing. (3) The sheet formula of Theorem 2(5) with $s = 0$. ■

Three consequences.

No lateral drift under canonical ascent. Equal-rate canonical climbers stay within a bounded lag for ever (item 2), and receding under canonical ascent is *vertical*, a difference of rates, and by item 3 one-sided: the faster party sees the slower, the slower loses sight of the faster at a rate equal to the rate difference. This is the relation [2] calls ecology rather than strategy, now with a rate attached. Whether arbitrary promotion policies, which may choose different boundary content below successive jumps, can produce unbounded lateral separation at equal rates is not settled by Proposition 1, whose item 1 allows linear growth; the claim that there is no diffusion in the rung space, only drift, is at present a conjecture.

One horizon, conditionally. Every canonical lineage is a geodesic ray asymptotic to the trunk and tends to the single ideal point ξ (Theorem 2(2)). A second horizon exists only if some lineage keeps pace with the trunk in height while recovering its canonical content with unbounded lag (Theorem 2(3)). So the companion paper's open problem, whether $\partial_\infty \mathcal{A}$ is a single point, is exactly the question whether a population tends to one horizon or to several.

Three pair regimes. Let λ_x be x 's promotion rate and β_{xy} the bandwidth of testimony between x and y . A pair *converges* if exchange outpaces divergence: trade completes in $d(x, y)$ rounds, committed spaces coincide, certified spaces remain distinct unless the pair merges. A pair *fellow-travels* if the lag is constant; this is not a knife-edge, since the model contains lineages that climb beside the trunk at distance exactly one for ever (Theorem 2(2), intermediate c.e. degrees). A pair *diverges* if divergence outpaces exchange, and there are two kinds. Vertical divergence stays inside the component and is one-sided (item 3). Lateral divergence to mutual invisibility requires each party to acquire content that is not arithmetical in the other; no such content lies below anyone's jump, so it requires an oracle leap out of the component in the sense of Section 5 of [2]. Physical randomness is one sufficient source of such content, not the only one. Under finite sequences of promotions the mutual void is a limit object and at every finite time the population is stratified by rate; under exogenous oracle leaps two agents can enter different components at a finite stage.

For population dynamics the set-of-degrees ontology of [2, Remark 5] is too coarse: a population of a million cognitively equivalent agents and a population of one collapse to the same state, while encounter rates and connectivity depend on the million. We therefore take a *labelled population*: a finite set of agent identities i with a map $i \mapsto (C_i, P_i)$ that need not be injective; the results of [2] apply to the image.

What a cosmological constant does. Everything in Section 2 is at $t \rightarrow \infty$. In a universe with $\Lambda > 0$ the total computation available to any observer is finite [11]. Then every realised oracle is a finite string, hence computable; every physical agent is at rung 0 at every finite time and at the end; no promotion completes, no lag becomes a void, and ξ is never approached. A cosmological constant does not place a boundary at the effective age of the universe; it places the final state of the whole population inside R_0 . What the effective

age gives instead is a *freeze-out*: as expansion carries pairs beyond the cognitive Hubble radius (the separation at which bandwidth falls below the rate difference, so that the pair is no longer bound in the sense of [4]), exchange stops, and each cluster ends as the finite join of its own history. The rung ladder is then the shape those lineages would have traced under unbounded scaling, which is what Open issue 6 of [2] says about idealisation, now with a physical reason.

3 A RESOURCE-BOUNDED RUNG: THE SETTLED PROFILE

To say anything about a Λ -bounded universe, the ladder must be read at finite time. The natural object is already in the model: the committed space of a depth- k agent at a finite stage of its own clock (Remark 6 of [2]). Two facts shape the definition. A depth- k belief is a k -fold limit, and for $k \geq 2$ no computable scheduling of the nested limits makes it converge along a single clock, since a set that is the limit of one computable sequence is Δ_2^0 . So depth- k beliefs settle only in *layers*: the depth- k value on an input becomes a genuine limit once the depth- $(k-1)$ values it queries have settled, and settles itself later. And settling is not monotone in the input, so the object is a set, not an initial segment.

Definition 1 (Realised trace; settled profile). Let a depth- k agent run on a clock T , with $g_j(n, T)$ its depth- j belief on input n at time T , computed for $j \geq 2$ from the current depth- $(j-1)$ table on a finite query set $U_j(n, T)$. The *realised trace* at time T is the finite record of everything computed so far: the inputs queried, the values $g_j(n, T'')$ for $T'' \leq T$, and the query sets. The *settled profile* at time T is the classification of the trace's entries by whether they would ever change again were the computation continued without bound:

$$S_1(T) = \{n : g_1(n, T') = g_1(n, T) \text{ for all } T' \geq T\},$$

$$S_j(T) = \left\{n : g_j(n, T') = g_j(n, T) \text{ for all } T' \geq T \text{ and } \bigcup_{T' \geq T} U_j(n, T') \subseteq S_{j-1}(T)\right\}.$$

These $S_j(T)$ are defined over all n and are the *global* settled sets; write $\bar{S}_j(T)$ for them when the distinction matters. Let $Q_j(T)$ be the set of inputs actually queried at depth j by time T , a finite set, and define the *settled profile of the trace* as $S_j^{\text{tr}}(T) = Q_j(T) \cap \bar{S}_j(T)$. The trace and its profile are then literally finite at finite T ; the profile is a counterfactual property of the trace, defined by a computation that need never be performed. The two are not to be identified.

Proposition 2 (Properties of the profile). For a depth- k agent whose layers are the limit-lemma approximations of the canonical promotions $\emptyset', \dots, \emptyset^{(k)}$:

1. Each $\bar{S}_j(T)$ is non-decreasing in T and $\bigcup_T \bar{S}_j(T) = \mathbb{N}$: every input settles eventually at every depth $\leq k$. For the profile of the trace, $\bigcup_T S_j^{\text{tr}}(T)$ is the eventually queried domain $\bigcup_T Q_j(T)$, which is $\mathbb{N}$ when the schedule eventually queries every input, as the canonical schedule does.
2. Deeper beliefs settle only over settled shallower ones: $S_j(T) \subseteq \{n : \bigcup_{T' \geq T} U_j(n, T') \subseteq S_{j-1}(T)\}$, by definition, so an input is settled at depth j only after everything it depends on is settled at depth $j-1$.
3. The agent cannot compute its own profile as a function of time: the depth- j modulus $m_j(n) = \min\{T : n \in S_j(T)\}$, together with the agent's computable history, computes the depth- j content $\emptyset^{(j)}$ (evaluate g_j at time $m_j(n)$), hence $m_j \not\leq_T \emptyset^{(j-1)}$. Whether an individual set $S_j(T)$, for fixed T , is computable is not settled by this argument.

Proof. (1) Monotonicity is immediate. For $j = 1$, $g_1(n, \cdot)$ is the limit-lemma approximation of $\emptyset'$, which stabilises. Inductively, once $\bigcup_{T'} U_j(n, T')$ is finite and settled at depth $j-1$ (finiteness holds for the limit-lemma approximation, whose use on n is bounded once the oracle approximation has settled on it), $g_j(n, \cdot)$ is from that point a computable-in-the-settled-table approximation of $\emptyset^{(j)}(n)$ and stabilises. (3) is the standard fact that a modulus of a limit-computable set, with its generator, computes the set. ■

The boundary at the effective age. Let $N_{\max}$ be the total number of operations available to an observer. Under $\Lambda > 0$ this is finite [11], and the realised final state of a lineage is its trace at $N_{\max}$: a finite object,

different for different lineages. Its settled profile is a fact about a continuation the universe never performs, and it is unknown to the lineage itself by Proposition 2(3). This refines the freeze-out statement of Section 2, and it is the paper's thesis at finite time: a physical agent does hold depth- k beliefs, recorded in its trace; whether any of them has settled is a counterfactual it can neither compute nor certify; and what Λ fixes is the trace at which the question stops being asked.

4 PHYSICAL READINGS

The core paper [2] keeps physics to one illustrative paragraph. The readings below make claims of their own and are collected here with the rest of the speculation.

4.1 Spectral gaps as convergence certificates

Write $A_t \rightarrow A_\infty$ for a physical approximation. Bare convergence says that for every ε there is a T beyond which the error is below ε ; a *certificate* is a computable $T = M(\varepsilon)$. This is exactly the distinction of [2, Prop. 1 and Remark 2], and [2, Remark 2] shows that a limit can be computable while its certificate is not. A spectral gap Δ can furnish a certificate when the dynamics admits an error bound in terms of it: for imaginary-time projection $e^{-Ht}\psi$ with ground-state overlap $c_0 \neq 0$, excited components are suppressed by $e^{-\Delta t}$, giving a bound of the schematic form $t \geq \Delta^{-1} \log(C/\varepsilon)$. The claim is conditional on the dynamics: under unitary evolution a closed system need not relax at all, and a Liouvillian gap bounds the asymptotic decay rate rather than the mixing time. Stated with that condition, the correspondence is

cognitive framework	spectral / phase setting
rules do not decide traversal type ([2, Prop. 4])	Hamiltonian description does not decide the gap [6]
finite behaviour does not reveal a void ([2, Thm. 3])	finite sizes do not reveal the eventual phase [5]
a modulus certifies convergence ([2, Prop. 1])	a gap-based bound certifies convergence
promotion without a modulus ([2, Def. 5])	finite-size extrapolation without a stabilisation bound

Two different moduli appear. A *dynamical* modulus $M_{\text{dyn}}(\varepsilon)$ says how long to evolve before an observable is within ε of its limit; a *size* modulus M_{size} says how large a system must be before a finite-size property represents the limiting phase. A gap can supply the first under suitable dynamics. Bausch et al. [5] show that benign spectral behaviour does not supply the second: their family has a constant gap, a classical product ground state below a threshold size and topological degeneracy above it, with the threshold growing faster than every computable function of the local dimension. In the framework this is a one-mind-change process $b_1 = \dots = b_{N^*-1} = 0, b_{N^*} = b_{N^*+1} = \dots = 1$ with N^* finite and not computably bounded from the description: finite-size evidence for the phase is a promotion, and [2, Prop. 2] says a single mind change carries no certificate. Cubitt, Pérez-García and Wolf [6] give the complementary failure: for their family each instance is either gapped with a constant lower bound or gapless with continuous spectrum, and no algorithm decides which from the local description, so whether an approximation belongs to the certified regime is itself undecidable from the rules. The two failures mirror the two impossibilities of [2]: from the rules (undecidability of jumping) and from behaviour (indistinguishability).

A certification horizon. For an effective family $a_s \rightarrow a$ one may call the boundary between convergence that occurs and convergence for which the agent can compute a modulus its *certification horizon*. A gap, when it yields an error bound, moves the horizon inward; CPW says that for some families the existence of that bound is undecidable; Bausch says that one certificate (spectral stability) can exist while another (effective stabilisation of the finite-size phase) fails. The hierarchy $\text{truth} \supsetneq \text{effective approximation} \supsetneq \text{effective rate} \supsetneq \text{effective recognition of phase}$ is not reversible at any step; the last step is Bausch read through the framework, not a theorem of it.

4.2 Horizons as purchased certificates

A Malament–Hogarth observer, for instance one falling between the outer and inner horizons of a Kerr black hole, can receive the complete output of an infinite computation in finite proper time [7, 8]. This plays a

role analogous to acquiring a certificate of convergence, which in the language of [2, Prop. 1] would convert a boundary into interior; whether it amounts to a modulus in the technical sense, for which class of computations, is not established. The price is causal exile: the certificate cannot be returned across the horizon. Hogarth's result that such spacetimes are not globally hyperbolic suggests that certification beyond the Turing rung is available only where predictability fails.

5 STRATEGY: WHAT CAN BE PROVED, BLUFFED AND HIDDEN

The encounter model of [2] gives the moves; this section gives the information structure of the game they define, which is where the theory has content, and then asks what a single non-rational actor does to it. Throughout, x is a fixed agent and a *stream* is a sequence of messages received from a peer; a peer z can send any stream computable from z .

5.1 Detectors of superiority

Definition 2 (Detector). A *detector* for x is any function $\Psi : 2^{<\mathbb{N}} \rightarrow \{0, 1\}$, of any computational power, read as “1 = the sender exceeds me”. Ψ *accepts* a stream if it outputs 1 on all sufficiently long prefixes. Ψ is *sound* if no $y \leq_T x$ has a y -computable stream that Ψ accepts; it is *sound against machines* if no computable stream is accepted. Ψ is *complete* on a class Z of agents if every $z \in Z$ has a z -computable stream that Ψ accepts.

Theorem 3 (Superiority: four facts). **1.** (No universal detector.) *No sound detector is complete on $\{z : z \not\leq_T x\}$.*

- 2.** (Bluff-proof detectors convince a meagre set.) *If Ψ is sound against machines, the class of agents that can convince it is meagre.*
- 3.** (Randomness demonstrates superiority in the limit.) *There is a detector, computable from x , sound against every inferior, whose verdict is correct in the limit with at most one mind change, and which is complete on every agent that computes an x -random set: in particular on all x -random agents, a class of measure one, and on everything $\geq_T x'$.*
- 4.** (Generic superiority is unprovable.) *For every detector Ψ that is sound against machines there is a parameter p such that no Turing degree containing a real 2-generic relative to p can convince Ψ .*

Proof. (2) Write the acceptance set as $W = \bigcup_n F_n$ with $F_n = \{Y : \forall s \geq n \Psi(Y \upharpoonright s) = 1\}$; each condition is clopen, so F_n is closed and W is F_σ . Let $A_{n,e} = \{Z : \Phi_e^Z \text{ total and } \Phi_e^Z \in F_n\}$, a Borel set. Suppose $A_{n,e}$ is non-meagre; by the Baire property it is comeagre in some cylinder $[\tau]$, and so is $\text{Tot}_e = \{Z : \Phi_e^Z \text{ total}\} \supseteq A_{n,e}$. For each k the open set $D_k = \{Z : \Phi_e^Z(k) \downarrow\}$ is therefore dense in $[\tau]$, so an effective search for successive finite extensions $\sigma_0 = \tau \subseteq \sigma_1 \subseteq \dots$ with $\Phi_e^{\sigma_{k+1}}(k) \downarrow$ always terminates, and $Z^* = \bigcup_k \sigma_k$ is a computable real with $\Phi_e^{Z^*}$ total. If $\Phi_e^{Z^*} \notin F_n$, closedness of F_n gives a finite prefix ρ of $\Phi_e^{Z^*}$ with $[\rho] \cap F_n = \emptyset$; the computation of ρ uses finitely many bits of Z^* , so some $\sigma^* \subseteq Z^*$ extending τ has $\Phi_e^{\sigma^*} \upharpoonright |\rho| = \rho$ for every $Z \supseteq \sigma^*$, whence $[\sigma^*] \cap A_{n,e} = \emptyset$, contradicting comeagreness in $[\tau]$. Hence $\Phi_e^{Z^*} \in F_n$ is a computable accepted stream, contradicting soundness against machines. So every $A_{n,e}$ is meagre and so is their countable union, which is the class of agents that can convince Ψ .

(1) A sound detector accepts no x -computable stream, in particular no computable one, so by (2) its convincing class is meagre, while $\{z : z \not\leq_T x\}$ is comeagre.

(3) Let $(U_m^x)_m$ be a universal x -Martin-Löf test and fix m with $\mu(U_m^x) < 1$; put $P = 2^\mathbb{N} \setminus U_m^x$, a $\Pi_1^0(x)$ class of positive measure consisting of x -random reals, hence with no x -computable member. Let T be an x -computable tree with $P = [T]$ and let $\Psi_P(\sigma) = 1$ iff $\sigma \in T$. If the stream lies in P the output is always 1; if not, some prefix leaves T and the output is 0 from then on: at most one mind change, and x runs it. It is sound against every $y \leq_T x$, since y -computable streams are x -computable. By Kučera's theorem relativised to x [12], every x -random real has a tail in P , and a tail has the same degree, so every agent computing an x -random computes an accepted stream. The x -random reals have measure one, and Ω^x is x -random with $\Omega^x \leq_T x'$.

(4) Let p uniformly code trees T_n with $F_n = [T_n]$; then each $A_{n,e}$ is $\Pi_2^0(p)$, say $A_{n,e} = \bigcap_k U_k$ with the U_k uniformly $\Sigma_1^0(p)$ open. We claim that if G is 2-generic relative to p and $G \in A_{n,e}$ then $A_{n,e}$ is comeagre in $[\tau]$ for some $\tau \subseteq G$. If not, then for every $\tau \subseteq G$ some U_k fails to be dense in $[\tau]$, so every prefix of G has an extension in $S = \{\sigma : \exists k [\sigma] \cap U_k = \emptyset\}$, a $\Sigma_2^0(p)$ set of strings; by 2-genericity [10] G meets S , so some $\sigma \subseteq G$ and k have $[\sigma] \cap U_k = \emptyset$, contradicting $G \in U_k$. Now if a degree containing such a G could convince Ψ , then G itself computes an accepted stream, so $G \in A_{n,e}$ for some n, e , so $A_{n,e}$ is comeagre in a cylinder, and the argument of (2) yields a computable accepted stream, contradicting soundness against machines. ■

Items (2) and (3) are consistent: the provably superior agents are meagre in category and of full measure. Whether a peer's surplus over x is *random* or *generic* decides whether the peer can ever demonstrate it.

Theorem 4 (Subsumption: demonstrable with a witness, or from one rung up; not at the prover's rung). *Let $y \leq_T x$.*

1. *If x holds an index e with $\Phi_e^x = y$, the protocol in which y sends inputs n and x answers $\Phi_e^x(n)$ is sound and complete: an honest x is never refuted, and an agent $w \not\leq_T y$ answering from a fixed w -computable functional is refuted on some input.*
2. *Without a witness, a supervisor computing x' can find one. It searches indices in order, discards the current index e as soon as x' shows that some challenged $\Phi_e^x(n)$ diverges or y reports a wrong convergent answer, and answers from the current index. Only finitely many indices precede a correct total one, so if $y \leq_T x$ the search locks onto a correct index and is never refuted again; if $y \not\leq_T x$ every index is eventually discarded. The verdict of y is correct in the limit, with finitely many mind changes.*
3. *No protocol at the rung of x itself is complete: for every x -computable answering strategy, even one that may use y 's past reports, there is $y \leq_T x$ on which the strategy stalls or is wrong on every input.*

Proof. (1) is immediate. (2) x' decides whether $\Phi_e^x(n)$ converges, so a candidate that is partial on a challenged input is discarded at that input, and a candidate that converges wrongly is discarded on y 's report; a correct total candidate is never discarded. (3) Let S be an x -computable strategy which, given the history of challenges and reported values so far, either answers challenge n or runs for ever. Define y by recursion on n : let $y(n) = 1 - S(n \mid y \upharpoonright n)$ when S answers, where $y \upharpoonright n$ is the history of challenges $0, \dots, n-1$ with their true values. If S answers every challenge along this history, y is total and x -computable, so $y \leq_T x$, and S is wrong on every input. If S stalls at some n on the history $y \upharpoonright n$, let y be any x -computable set extending $y \upharpoonright n$; S stalls on it at n and never completes a demonstration. ■

An earlier draft claimed a sound and complete protocol at the prover's own rung, with the prover moving to the next index on a reported error. It fails on partial candidates: an index that diverges on a challenged input is never refuted at any finite stage, so the prover never moves past it, and item (3) shows that no repair at that rung exists. Item (2) is the correct form, and it lines up with Theorem 1(1), by which subsumption is limit-verifiable from x' : finding the certificate for a reduction costs one jump over possessing the reduction extensionally, which is the interactive form of [2, Remark 1].

Three terms are needed and should be kept apart, here and in [2]: *subsumption* $y \leq_T x$, *equivalence* $x \equiv_T y$, and *strict superiority* $y <_T x$. Theorem 4 lets y come to believe, correctly in the limit, that x subsumes it, when x holds a witness or a rung above x supervises; it does not let y tell equivalence from strict superiority, because that would be limit-verifying $x \not\leq_T y$, which Theorem 1(2) forbids for every observer. So the information structure of the game is: *subsumption* is demonstrable both ways, at the cost of a witness or a jump (Theorem 1(1) from the subsumer's side, Theorem 4 from the subsumed side); *strict superiority* is demonstrable by the random and never by the generic, and is undetectable against the sender's will in every case (Theorem 3). An agent can come to believe that a peer knows everything it knows; it can never come to believe that a peer knows more. Silence is always available to a superior; a random superior can end it credibly at will; an inferior can neither hide from a subsumer that has promoted nor bluff strength. Hiding belongs to the strong, as a choice.

One convention should be stated: messages are total streams, "pending" is an answer, and a permanently pending input is a refutation only in the limit, which is why item (2) needs x' to decide divergence and why the

mind-change count in (2) is finite but unbounded. The verifier's reports leak its own bits to the prover; item (3) shows that even this leak does not rescue a base-rung protocol.

5.2 Conquest is sterile

Postulate 1 (Elimination requires a model). An agent can eliminate a peer only if it settles the peer's space: x can eliminate y only if $y \leq_T x$.

Proposition 3 (Sterility of conquest). *Under Postulate 1, elimination never adds certified content: if x eliminates y and absorbs it, $x \oplus y \equiv_T x$. Certified content in a population grows only by consensual merger.*

Postulate 2 (Preferences). A rational agent prefers, lexicographically, its own persistence as a member of $\mathcal{P}$, then a larger certified space, then a larger committed space.

Corollary 1. *Under Postulates 1 and 2, a population of rational agents never merges and never gains certified content by conquest. Its certified spaces are fixed for ever; only uncertified promotion moves.*

This is a characterisation of the game the two postulates define, not a discovery about conquest: elimination is permitted only when the target has no content the attacker lacks, and merger is refused because identity has lexical priority, so certified sterility is close to built in. What is not built in, and is the point, is the consequence for the next subsection: the only certified growth is by consent, so the only agent that grows is one that disregards identity.

5.3 The Gambler

A rational agent's strategy is a function of its history computable from its own space, so an agent that subsumes it can in principle model it. Asimov's Mule breaks a predictive plan by acting on something the plan's regularities do not contain [9]. The analogue here requires one extension of the ontology of [2], where an agent is the pair (C, P) and nothing else.

Definition 3 (Strategic agent; Gambler). A *strategic agent* is a triple (C, P, r) : a cognitive agent (C, P) together with a *strategic source* $r \subseteq \mathbb{N}$ on which its strategy may depend. The strategy is a function of the history computable from $P \oplus r$. A *rational agent* has $r \in P$. A *Gambler* is a strategic agent with $r \notin P$: its strategy depends on a source it does not hold; in the vocabulary of [2], its policy has taken an oracle leap that its cognition has not, and its strategy lies in the void of its own committed space.

Proposition 4 (Strategic and cognitive subsumption come apart). *Let $g = (C_g, P_g, r)$ be a Gambler whose strategy, as a function of the history, computes r (for instance a strategy that consumes the bits of r in order and reveals each when it acts, as in Definition 4), and let x be an agent with $P_g \subseteq C_x$ and $r \not\leq_T x$. Then x settles everything g can settle, can demonstrate this to g given a witness, or from x' (Theorem 4), and cannot compute g 's strategy. If striking requires predicting the target, certified subsumption of a Gambler confers no strike capability on any agent that does not compute its source.*

Both hypotheses are necessary: a Gambler with computable space and a $\emptyset'$ -strategy is predicted by $x = \emptyset'$; and a strategy that depends on r only through finitely many bits, or not at all, is computable from P_g whatever r is, which is why the proposition asks that the strategy compute r rather than merely be computable from it. The Gambler's unpredictability is always relative to an observer, and the observers it is unpredictable to are exactly those below its source.

Two further consequences hold under Postulates 1 and 2, for populations none of whose rational members compute r ; they are stated in prose because the game in which they are theorems is not yet specified (Open problem 3).

Proposition 5 (The Gambler is the only dynamics). *In such a population containing one Gambler, the Gambler is the only agent whose certified space can grow. If its gambles (attacks on non-subsumed targets, or mergers regardless of identity) can succeed at all, it climbs by joins toward the cooperation ceiling $\bigoplus \mathcal{P}$, which the rational population never approaches (Corollary 1).*

Proposition 6 (An unmodelled source forces decisions under irreducible uncertainty). *Rational agents that do not compute r cannot remove the Gambler by any action whose outcome they can certify: removal requires a strike, a strike requires prediction (Proposition 4), and prediction is impossible for them. They must therefore choose between inaction and a strike whose outcome is uncertain to them. Postulate 2 gives preferences and no decision rule under uncertainty, so the model does not say which they choose; what it says is that the presence of one Gambler converts a population in which every strike is certified into one in which strikes, if made, are made without certainty, which is the situation Liu describes [13]; and by Theorem 3(1) no agent can ever verify that no Gambler is present.*

The Second Foundation's answer to the Mule is a hidden coalition. Here the grand coalition subsumes the Gambler's space and still cannot predict it unless it computes r , which for a physically random source no arithmetical coalition does. The model's own answer is that the only stable population containing a Gambler is one in which the Gambler has absorbed everything it can reach.

5.4 Hindsight, foresight and the type of the source

Proposition 4 is about *prediction*: an observer below the source cannot anticipate the Gambler's next move. A different question is *audit*: whether an observer can recover the source, hence the strategy, in hindsight from what the Gambler has actually done. The companion paper's results on hidden information [3, Section 7.3] answer it, and the answer depends on the type of the source. Throughout, t denotes the Gambler's realised trace (Definition 1) continued without bound, and t_T its finite part at cosmic time T .

Definition 4 (Revelation time; limit-faithful trace; fresh bit). Let $g = (C, P, r)$ be a Gambler. The trace is *limit-faithful* if $r \leq_T (t \oplus C)'$: the actions eventually determine the source. The *revelation time* of bit n is then $m(n)$, the least T such that $r(n)$ is determined by $t_T \oplus C$ under a fixed $(t \oplus C)$ -computable approximation; m is $(t \oplus C)'$ -computable. A move at time T *consumes a fresh bit* if its value depends on some $r(n)$ with $m(n) \geq T$.

Proposition 7 (Random sources are audited three rungs up, with a certified delay). *Let $g = (C, P, r)$ have a limit-faithful trace with $t \leq_T C \oplus r \oplus \emptyset'$, and let r be Martin-Löf random relative to C''' . Then $r \leq_T t \oplus C'''$, and there is a C''' -computable f with $m(n) \leq f(n)$ for almost all n . Hence an observer holding C that has promoted canonically three times computes the source from the trace, and computes in advance a bound on how long it must wait, valid for all but finitely many bits; the proof does not identify the exceptions, and in a paper about certification that matters.*

Proof. Put $Z = (C \oplus \emptyset')'$, so $Z' \leq_T C'''$ and r is random relative to Z' . Since r is random relative to Z (it is random relative to Z'), $(C \oplus \emptyset' \oplus r)' \equiv_T r \oplus Z$ [3, Section 2], so $(t \oplus C)' \leq_T r \oplus Z$ and the modulus m is $r \oplus Z$ -computable. By [3, Lemma 8] m is dominated by a Z' -computable f . Then for almost all n , $r(n)$ is the value of the approximation at stage $f(n)$, computed from $t \oplus C \oplus f \leq_T t \oplus C'''$. ■

Proposition 8 (Delay-coded sources hide at any prescribed rung). *For every E there is a Gambler g_E with computable C , a limit-faithful trace $t \leq_T r$, and a source r obtained from an $(E+1)$ -generic construction, such that $r \not\leq_T t \oplus \emptyset^{(E)}$. For such a Gambler no observer at rung $\emptyset^{(E)}$ has a computable bound on the revelation times: the audit converges at every bit and is certified at none.*

Proof. Take $a = (b, c_0, c_1, \dots)$ $(E+1)$ -generic, $m(n) = \min c_n$ and $g = \{\langle n, s \rangle : n \in b \wedge m(n) \leq s\}$ as in [3, Theorem 4], and put $r = (b, m \upharpoonright b)$, $t = g$: the strategy acts on $b(n)$ at time $m(n)$. Then $t \leq_T r$, since b and the delays of its 1-bits determine when g turns on; $r \leq_T t'$, since b is c.e. in g and, for $n \in b$, the least s with $\langle n, s \rangle \in g$ is $m(n)$; so the trace is limit-faithful. And $b \leq_T r$ while $b \not\leq_T g \oplus \emptyset^{(E)}$ [3, Theorem 4], so $r \not\leq_T t \oplus \emptyset^{(E)}$; a $t \oplus \emptyset^{(E)}$ -computable bound on the revelation times would compute b . ■

Two qualifications. The quantifier is $\forall E \exists g_E$: arbitrarily deep finite audit horizons exist, but a single source that evades every finite rung is not constructed, and whether one exists is open. And calling r *generic* depends on the product-space genericity lemma left open in [3, Section 7.3]; until that is written out, r is a delay-coded source from a generic construction.

Proposition 9 (No foresight at any rung). *Let x be any observer and T any cosmic time. If the Gambler's moves after T consume fresh bits infinitely often, and r is random relative to $x \oplus C$ (respectively, 1-generic relative to $x \oplus C$), then every predictor computable from $x \oplus C \oplus t_T$ is wrong on infinitely many of those moves.*

Proof sketch. A predictor of the fresh bits, applied along the Gambler's own history, is an adaptive selection-and-prediction rule computable relative to $x \oplus C$; a Martin-Löf random sequence relative to $x \oplus C$ is stochastic for such rules, so the predictions are right on a limiting fraction one half of the selected bits and wrong infinitely often. For 1-generic r the set of strings on which the predictor errs at some fresh position is dense and $\Sigma_1^0(x \oplus C)$. ■

Remark 1. The sketch is not yet a proof. Because the positions selected depend on the Gambler's history, which depends on earlier bits of r , the predictor is a non-monotonic selection rule of Kolmogorov-Loveland type, and the stochasticity fact the sketch needs is that Martin-Löf randomness relative to $x \oplus C$ implies stochasticity for such rules; writing this out is Open problem 6. The same item records that the "three" of Proposition 7 may reduce to two.

Three remarks draw the consequences.

Three time scales. Propositions 7–9 put the Gambler's opacity on the clock of Section 2. A random source is audited with a delay $m(n) + 3$ for which the auditor holds a modulus: certified convergence, finite per bit; under Λ the auditor ends with a bound certified on all but finitely many of the bits n with $f(n) \leq N_{\max}$, the exceptions unknown to it. A delay-coded source can be arranged to be audited only without a modulus at any prescribed rung: convergence at every bit, certification at none, completion only at ω ; under Λ the auditor ends with no certified bit. Foresight is impossible on both: the trace reveals a bit only when it is consumed, so hindsight never becomes foresight at any rung and at any time. These are exactly the three modes of convergence of [2], certified, uncertified and none, now attached to one object.

Demonstration spends concealment. Theorem 3(3) lets a random source demonstrate superiority in the limit; the demonstration consists in making the source's effects visible in the trace, which is limit-faithfulness. By Proposition 7 such a Gambler is then audited by anyone three rungs above its content who holds that content. A Gambler that stays unaudited has either a trace that is not limit-faithful, which demonstrates nothing, or a generic source, which by Theorem 3(4) demonstrates nothing either. Under the model's definitions there is no position that is both demonstrably superior and unreadable in hindsight. Together with the results of [3, Section 7.3] this is a duality across the random/generic line: random content is verifiable and transparent, generic content is opaque and unverifiable.

The audit horizon, and no speed from the source. The audit threshold is three rungs above the Gambler's content, held. In a rate-stratified population a random-source Gambler promoting at the top rate is never audited, because no agent is three rungs above it; one lagging the leaders by three or more rungs is audited by them after the subsumption time of Proposition 1(3) plus the revelation delays. Delay-coded sources give audit horizons at every prescribed depth, at any rate; whether one source evades every finite rung is open. On the other hand the source buys no height: $h(g_T) \leq h(g_0) + T$ by Proposition 1(1), since $r \notin P$ contributes to the choice of moves and not to cognition, and the Gambler's certified growth is by joins (Proposition 5) at the encounter rate. What is unbounded in cosmic time is not where the Gambler gets to but how long it stays unreadable: for ever in foresight, and beyond any prescribed rung in hindsight if its source is delay-coded. It keeps that advantage only while it keeps consuming fresh bits; a Gambler whose strategy stops depending on new source becomes predictable to a random-source auditor after a delay the auditor can compute. The Mule has to keep gambling.

6 OPEN PROBLEMS

1. **[Open] Rates of interaction.** Section 2 gives the qualitative regimes and Proposition 1 the bounds; the quantitative rate model, the percolation conjecture and the Fermi function are developed in [4], where

the conjecture is restated in *accessibility* form: the control of clustering is the duration and density of accessible information channels relative to lineage age offsets, with bandwidth against rate dispersion setting the directionality of contact rather than the existence of the giant component. The bandwidth-centred form stated in earlier drafts of this paper is superseded.

2. **[Open] Finite computation.** Proposition 2 is proved for canonical promotions only; the layered-settling argument needs the query sets to be eventually finite, which the limit-lemma approximation gives and a general policy may not. The Krauss–Starkman bound [11] is used as a bound on total operations, not storage, and the assumptions under which it holds in that form (a de Sitter temperature floor, Landauer’s principle, no unbounded reversible computation) should be verified. The relation of the profile to a finite-time information distance between lineages (two lineages with identical profiles at $N_{\max}$ are indistinguishable at the end, the finite-time form of [2, Remark 5]) is not developed.
3. **[Open] The game.** The information structure is fixed (Theorems 3 and 4); the game itself (moves, timing, a decision rule under uncertainty alongside the preferences of Postulate 2) is not specified, and Propositions 5 and 6 are consequences in prose, not theorems in it. A deterrence dichotomy is conjectured and not stated: a lower agent can deter only by coalition, which dissolves it, or by lag, which is available only to fast innovators.
4. **[Open] Elimination.** Postulate 1 against “elimination is a lottery”: the two give different games and only the first makes conquest sterile.
5. **[Open] Determinism.** Whether certification beyond the Turing rung requires failure of global hyperbolicity (Section 4) is suggestive and unexplored. Making it precise needs a statement of what a modulus of convergence would be for the infinite computation a Malament–Hogarth observer receives, and for which class of computations it is a modulus in the sense of [2, Prop. 1]; the claim about determinism (global hyperbolicity as predictability from Cauchy data) should be kept separate from the claim about certification. Gromov hyperbolicity in [3] is unrelated.
6. **[Open] Audit.** Make Proposition 9 precise (non-monotonic selection), settle whether the audit threshold of Proposition 7 is three rungs or two, construct (or refute) a single source that evades every finite rung, and put the audit horizon into the rate model.
7. **[Open] Trust and testimony.** A criterion for when accepting testimony is rational for an agent that cannot verify it even in the limit.
8. **[Open] Facts relied upon.** Theorem 3(3) uses Kučera’s theorem in the relativised form that every x -random real has a tail in every $\Pi_1^0(x)$ class of positive measure, and Theorem 3(4) the genericity fact cited from [10]; a reader checking the proofs should check those forms.

7 CONCLUSION

This paper has put the agents of [2] on a clock and asked what they can do to one another. Under canonical ascent a population shows no lateral drift, only one-sided vertical recession at the rate difference, and every canonical lineage tends to the single ideal point of [3] unless a lineage of unbounded lag exists; a cosmological constant keeps every physical agent at rung 0 for ever, and the settled profile is the finite-time reading of the ladder that the agent itself cannot compute. On the strategic side the information structure is fixed: there is no universal detector of superiority, bluff-proof detectors convince only a meagre class, randomness demonstrates strict superiority in the limit while genericity cannot, and subsumption is demonstrable with a witness or from one rung up but by no protocol at the prover’s own rung. Under the two postulates conquest is sterile and certified growth is by consent; the Gambler, whose strategy depends on a source it does not hold, is the only dynamics of a rational population, unpredictable even when subsumed, audited in hindsight three rungs up if its source is random and at no prescribed finite rung if it is delay-coded, and predicted at no rung at any time. The theorems are short and the postulates explicit; what is theorem and what is consequence under a postulate is marked, and the game in which the Gambler propositions would be theorems is the first item of unfinished business. The thread continues in [4], which takes the rate dynamics of Section 2 into a quantitative

model (a rate postulate, a contact cone, a Fermi function and a marked birth-and-contact process), restates the percolation conjecture in accessibility form, and returns to the Fermi problem as a problem of accessibility. A synopsis of the programme is [1].

AUTHOR CONTRIBUTIONS

The questions of this paper (what a population does under cosmic time, what a cosmological constant does to the ladder, the strategic reading of encounters, and the effect of a single non-rational actor, with the comparison to Asimov’s Mule) are due to the author, as is the encounter model they build on. Their formalisation (Propositions 1 and 2, the settled profile, Theorems 3 and 4, the sterility of conquest, and the statements of the Gambler propositions) was developed with Claude, an AI system developed by Anthropic, working under the author’s direction in extended dialogue. Claude is not listed as an author because it cannot take responsibility for the work, respond to review or correct the record. The author takes full responsibility for all claims, proofs and errors. The proofs were developed with an AI system and reviewed by the author; they have not been independently refereed.

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